



Building

Given a customer scenario, describe the appropriate development model (org-based vs package-based) and development environment (scratch org vs sandboxes).

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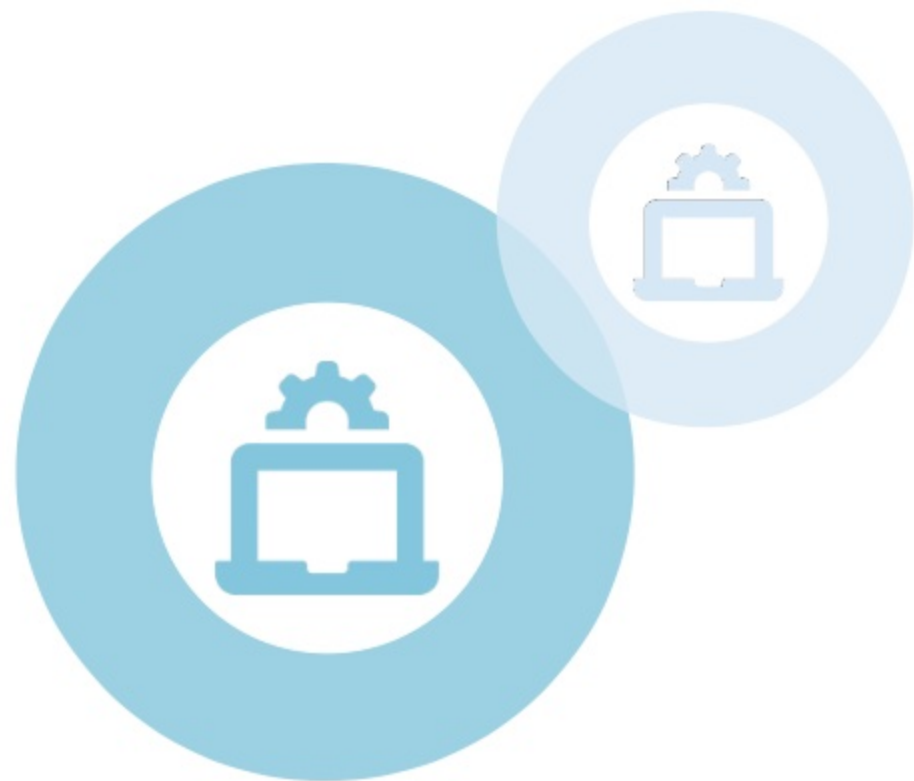
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After studying this topic, you should be able to:

-  Describe the main features of the org development and package development models.
-  Explain how scratch orgs, sandboxes, and development orgs are used for Salesforce development.
-  Identify the design factors that should be considered when choosing a Salesforce development model and a development environment.
-  Describe the appropriate development model and environment for the given business requirements.



Introduction

The **org development model** and **package development model** are two models that are often used for Salesforce development. The org development model leverages **sandboxes** and a **source control system** for development. On the other hand, in the package development model, metadata and code for applications and customizations are organized into **manageable packages**.

Scratch orgs are temporary orgs that can be spun up quickly for development. **Sandboxes** are copies of the production environment that are used for development, testing, and training. **Individual development orgs** and **partner development orgs** are also available for application development. Partner development orgs offer **higher API** and **storage limits** and **more licenses**.



Development Models & Development Environments

Org Development Model

The org development model leverages sandboxes for the development process and a source control system for managing changes.

Package Development Model

In the package development model, metadata and code for applications and customizations are organized into manageable packages.



Scratch Orgs & Sandboxes

A scratch org is a temporary org that can be spun up quickly for development. Sandboxes are copies of the production environment used for development, testing, and training.

Development Orgs

Individual and partner development orgs are free orgs available for development purposes. The partner development org offers higher storage and API limits and more licenses.

Development Models

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Org Development Model

The Org Development Model features **sandboxes** and **source control** for the development process.

❖ SANDBOXES

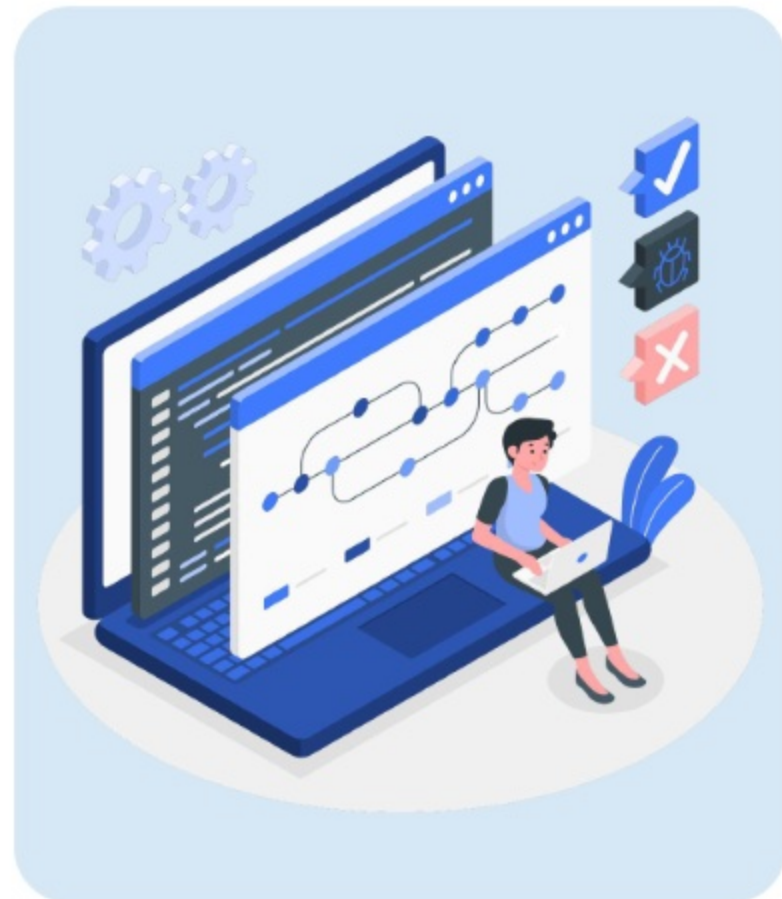
The **Org Development Model** leverages **sandboxes** at various stages of the development lifecycle, including **development**, **integration**, **testing**, and **training**.

❖ SOURCE CONTROL

A **source control system** is used to **manage changes** and ensure consistency across environments. A **central repository** is used to store all the code and metadata, which can also be stored on the **local machine**.

❖ SALESFORCE DX

Salesforce DX tools are used to **streamline the development workflow**.



Org Development Model

The Org Development Model features **sandboxes** and **source control** for the development process.

❖ TOOLS

Salesforce Extensions for VS Code and **Salesforce CLI** are used to enhance the development efficiency, from code retrieval to deployment. **Automated processes** can be created for testing and deployment.

❖ CHANGE MANAGEMENT

The development model ensures a robust **change management process** using tools for **tracking changes**, **managing deployments**, and facilitating **real-time collaboration** among team members.

❖ CHANGES

Changes must be tracked manually. Sets of changes are deployed to sandboxes and then to the production org.



Package Development Model

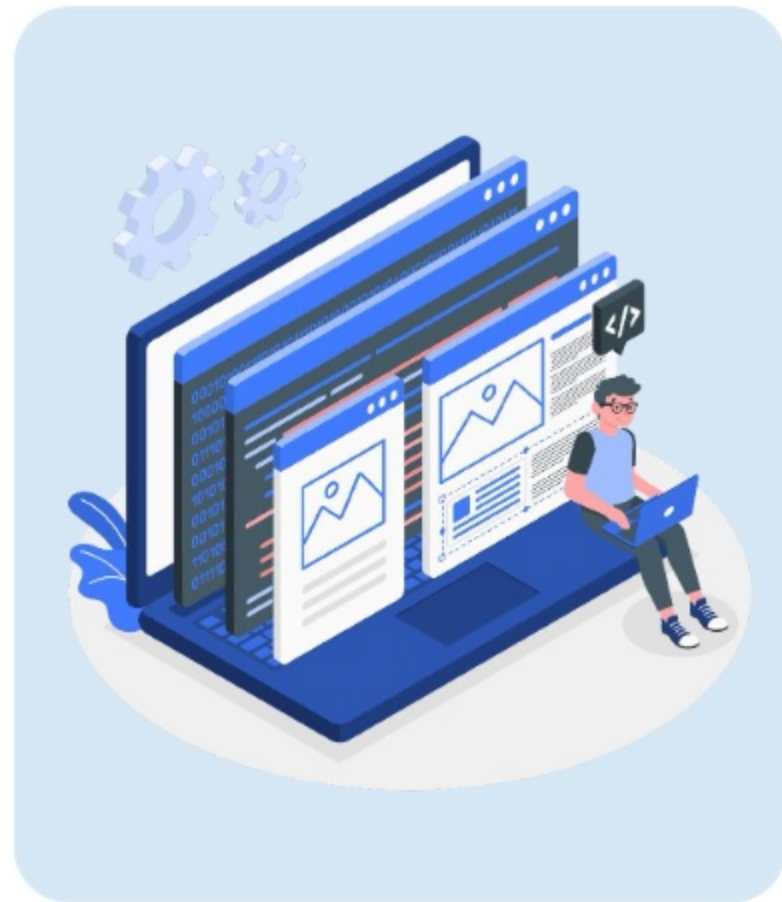
The Package Development Model uses **packages** to organize and manage components.

❖ MODULAR DEVELOPMENT

Metadata and code for applications and customizations are **organized into manageable packages** for modular development. This modularity supports **better organization** and **reuse of functionality** across **multiple projects** or **orgs**.

❖ VERSION CONTROL

Each **package version** is **immutable** once created, which allows **precise version tracking** and **control** over what is deployed. This facilitates **easier rollback** and **upgrades**.



Package Development Model

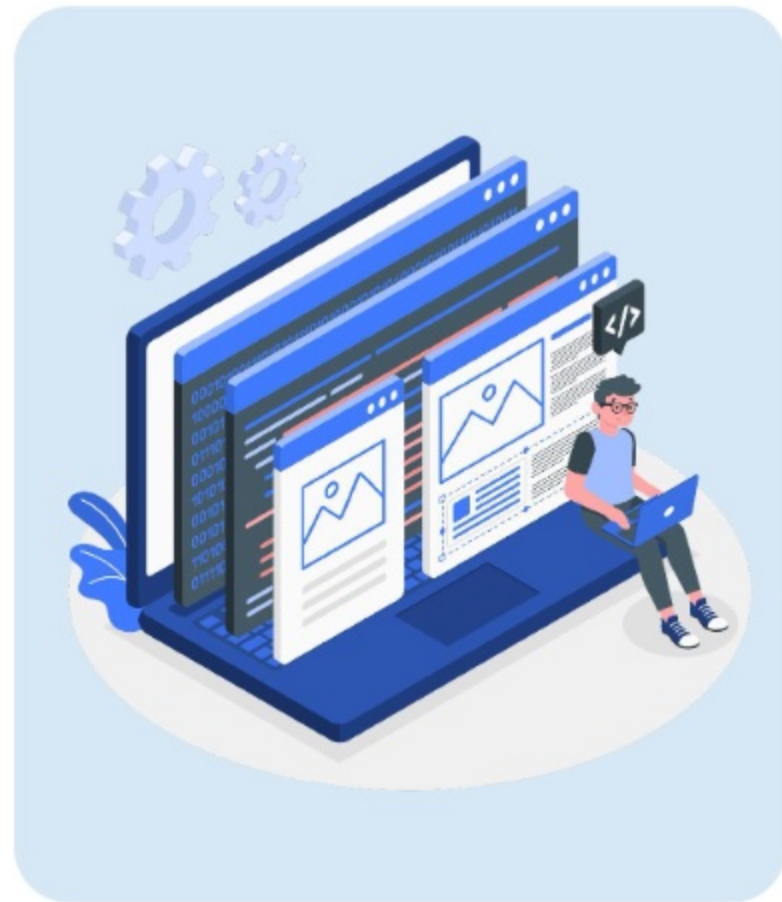
The Package Development Model uses **packages** to organize and manage components.

❖ DEPENDENCY MANAGEMENT

Packages can declare **dependencies on other packages**, enabling complex applications to be built from **smaller, interdependent modules**. This simplifies updates and maintenance.

❖ ISOLATION

Development in **isolated environments (scratch orgs)** is encouraged, minimizing the risk of conflicts and ensuring that development activities do not impact the production org directly.



Package Development Model

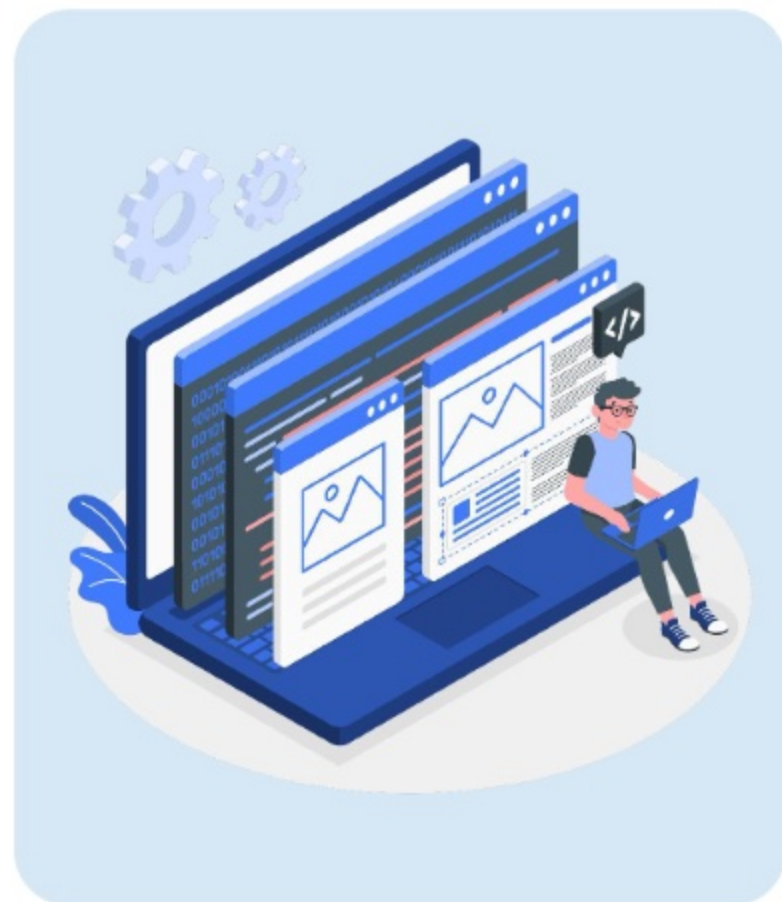
The Package Development Model uses **packages** to organize and manage components.

❖ SOURCE-DRIVEN DEVELOPMENT

The model uses **source tracking**, **source control**, and **continuous integration** and **deployment**. Changes are tracked on the local machine and in the default development environment.

❖ SOURCE TRACKING

Source tracking can be turned on in the source (production) org so that **Developer** and **Developer Pro** sandboxes automatically track changes between **sandboxes** created from it and **Salesforce DX projects**. It is **enabled by default** for **scratch orgs**. Only **metadata changes** that are **not yet supported by source tracking** need to be **tracked manually**.



Package Development Model

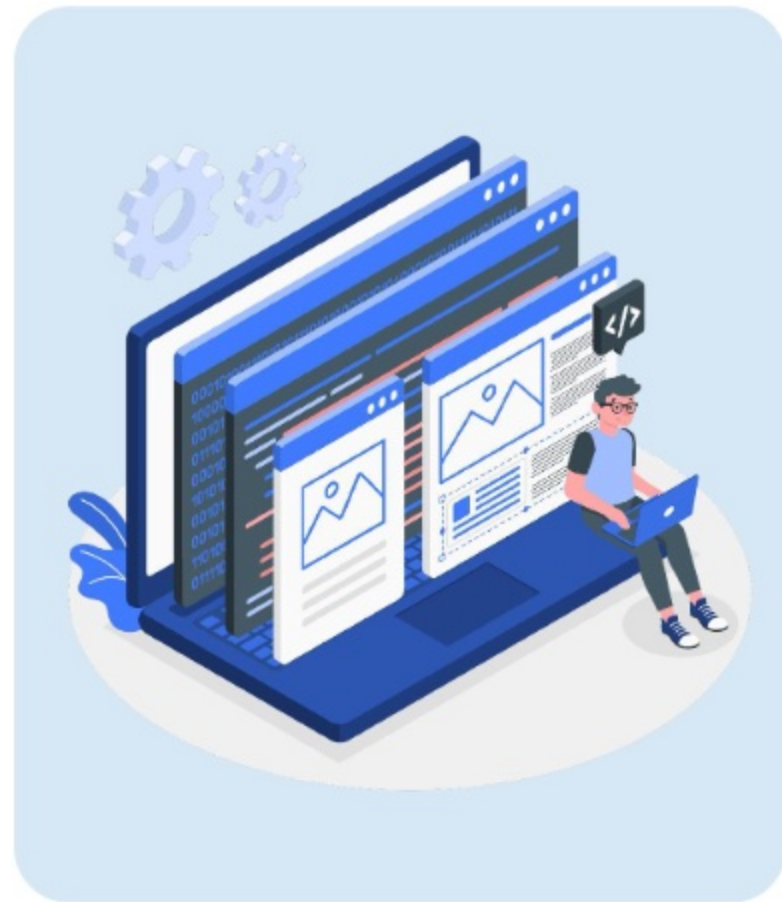
The Package Development Model uses **packages** to organize and manage components.

◆ DEV HUB

To create and manage **scratch orgs** and **packages**, **Dev Hub** must be enabled in the production org. To use packages, **Enable Unlocked Packages and Second-Generation Managed Packages** must be selected on the Dev Hub page in Setup.

◆ SALESFORCE DX

For development, a **Salesforce DX project** can be created. A **scratch org** can be created and set as the **default org** for development. Source can be **pushed** and **pulled** to keep the **local environment** and the **default development org** in sync.



Package Development Model

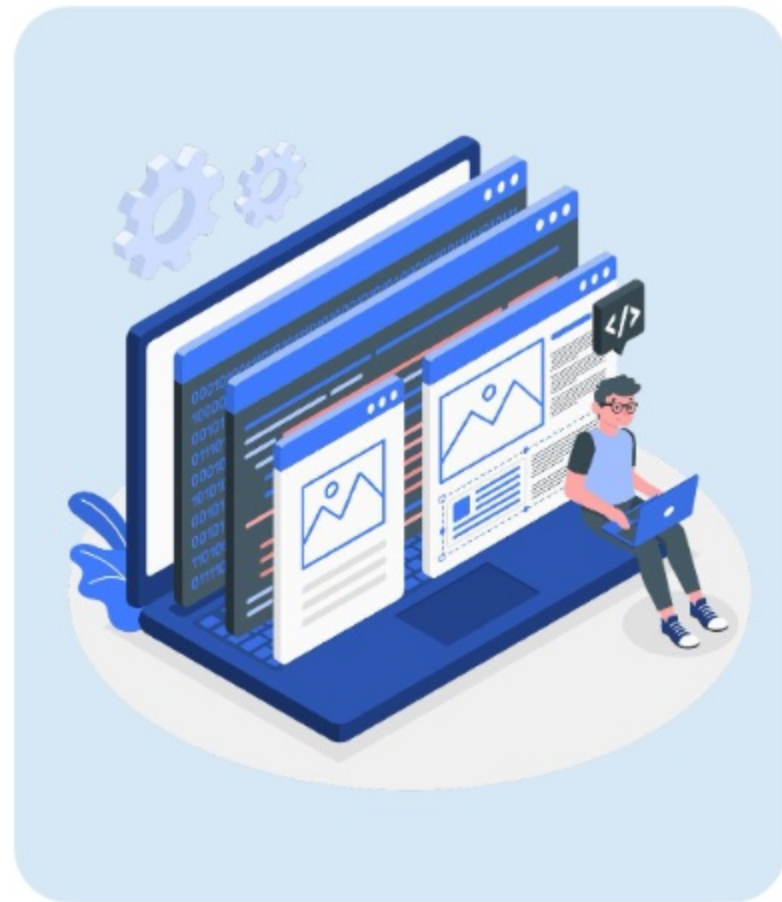
The Package Development Model uses **packages** to organize and manage components.

❖ SALESFORCE CLI

Salesforce CLI can be used to manage Salesforce DX projects, **create scratch orgs**, **push** and **pull metadata**, and **run unit tests**. It can also be used as part of **build automation scripts** to create and access environments, deploy source, install packages, and run tests.

❖ ADVANTAGES

The package development model offers various benefits, such as improved **team development** and **collaboration**, facilitating **automated testing** and **integration**, **greatly reducing the need for manual change tracking**, **more efficient** and **agile release cycles**, etc.



Development Environments

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Scratch Orgs

Scratch orgs are **temporary Salesforce orgs** that can be **quickly spun up** for development.

❖ TEMPORARY ORGS

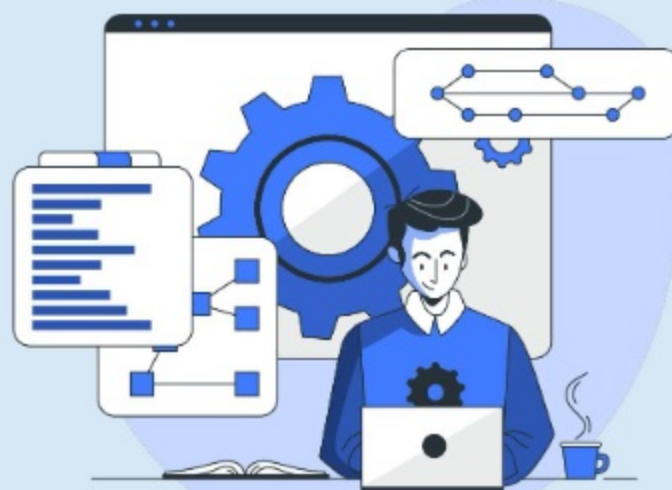
Scratch orgs are fully **configurable, disposable, and source-driven**. They allow developers to **emulate different Salesforce editions** with **different features and settings**.

❖ PACKAGES

Scratch orgs are ideal for the **development** and **testing of packages**, especially **unlocked packages**.

❖ AUTOMATION

Scratch orgs are suitable for **automated testing** and **continuous integration pipelines**.



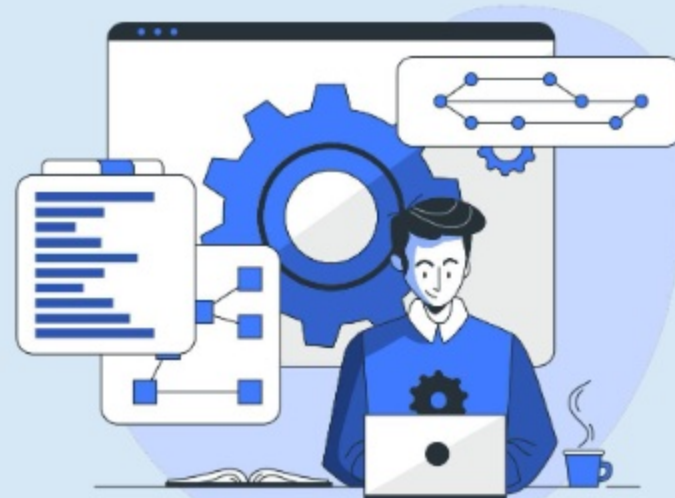
Scratch Orgs

Scratch orgs are **temporary Salesforce orgs** that can be **quickly spun up** for development.

❖ USE CASES

A scratch org can be spun up for the following:

1. Start a **new project**.
2. Start a **new feature branch**.
3. Test a **new feature**.
4. Start **automated testing**.
5. Perform **development tasks directly** in an org.
6. **Start from 'scratch'** with a fresh new org.



Sandboxes

Sandboxes are **copies of the production environment** that allow **development, testing, and training**.

❖ DEVELOPMENT

Sandboxes allow **isolated customization** and **development** work until changes are ready to be deployed to production.

❖ TESTING

Changes can be tested against **copies of the production data** and **users**.

❖ TRAINING

A sandbox can be used as a **training environment for end users**.

❖ COORDINATION

Individual changes can be coordinated into **one deployment to production**.



Developer Edition Org

An individual Developer Edition org is a **full-featured copy of Salesforce** that can be used indefinitely.

❖ USAGE

A Developer Edition org can be used to keep up with the **newest features**, build **custom apps**, and **test functionalities** before purchasing them. Partners can use it for **initial development** and **prototyping**.

❖ SIGN-UP

One can sign up for a **free Developer Edition org**. However, it **expires after 1 year of inactivity**.

❖ STORAGE & API

A Developer Edition org has a **storage limit of 20 MB** and **an API call limit (per 24 hours) of 15,000**.



Developer Edition Org

An individual Developer Edition org is a **full-featured copy of Salesforce** that can be used indefinitely.

❖ LICENSES & LIMITS

The org offers a **limited** number of user, feature and permission set **licenses**. It is **not possible to submit** feature/license/limit requests. For example, only **2 Salesforce user licenses** are available.

❖ MANAGED BETA TESTING

A Developer Edition org **supports managed beta testing**.



Partner Developer Edition Org

A Partner Developer Edition org is a **free org** available for **ISVs and consulting partners**.

❖ USAGE

A Partner Developer Edition org is typically used by partners for **application development** and **initial testing**.

❖ SIGN-UP

The **Environment Hub** is used to **create** and **log in** to a Partner Developer Edition org. However, it **expires after 1 year of inactivity**.

❖ STORAGE & API

A Developer Edition org has a **storage limit of 250 MB** and **an API call limit (per 24 hours) of 50,000**.



Partner Developer Edition Org

A Partner Developer Edition org is a **free org** available for **ISVs and consulting partners**.

❖ LICENSES & LIMITS

A Partner Developer Edition org offers a **higher** number of user, feature and permission set **licenses**, compared to an individual Developer Edition org. It is also **possible to submit** feature/license/limit requests. For example, **20 Salesforce user licenses** are available.

❖ MANAGED BETA TESTING

A Partner Developer Edition org **supports managed beta testing**.



Design Factors

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Choosing Development Model & Environment

Various **design factors** are necessary to consider for choosing an appropriate development model and environment.



PROJECT

Larger, more complex projects benefit from package-based development and the use of sandboxes or scratch orgs depending on the testing needs.



TEAM

Org-based development is suitable for **smaller teams**. Package-based development is appropriate for **larger, more collaborative teams**.



RELEASE

Package-based development with scratch orgs is ideal for **sophisticated release management** and **continuous integration**.



LONGEVITY

For **long-term projects** requiring **frequent updates** and **maintenance**, sandboxes should be considered for development for stability and ease of integration.

Business Scenario

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Business Scenario

Read the business scenario below and determine the most appropriate solutions for the company's requirements.



Cosmic Digital Frontiers, an innovative leader in virtual reality (VR) entertainment and educational platforms, develops immersive VR experiences for a global audience. Spanning markets in the Americas, Europe, and Asia, the company leverages Salesforce to streamline customer interactions, sales processes, and content delivery mechanisms. As the demand for VR content skyrockets, the company is at a crucial juncture, needing to efficiently manage the development of new features, address technical issues, and roll out updates to stay ahead in the highly competitive VR industry. The company seeks to implement a development strategy that not only accelerates innovation but also promotes effective collaboration among its globally dispersed teams. To achieve this, it requires a seasoned Salesforce architect to recommend the most suitable development model and environment that aligns with its vision for delivering unparalleled VR experiences.

Requirements & Solutions

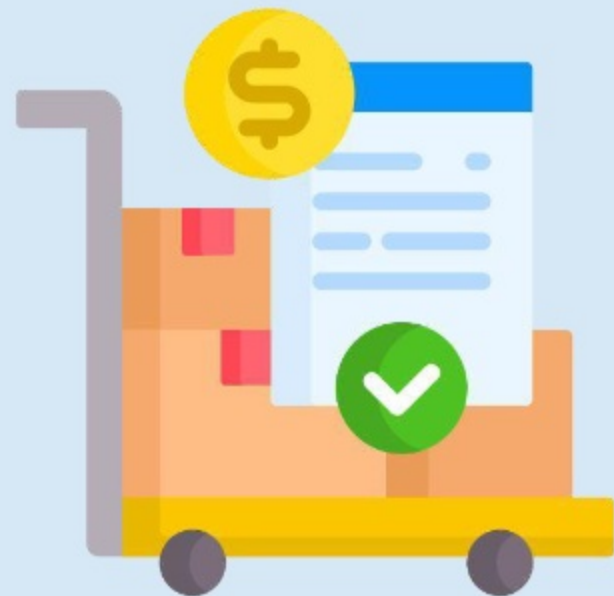
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Requirement 1



SCENARIO

Most of the Salesforce development teams of the company have been using the org development model for custom applications and customizations in Salesforce. A Sales Cloud development team has developed various extensions that are being used by internal users. The company recently acquired a software development company that develops extensions of an AppExchange app. The CTO would like both development teams to use the same Salesforce org to work on separate projects while following the same set of processes. The architect needs to recommend a suitable development model and environment.



Solution 1



SOLUTION

The development teams should adopt the **package development model** so that they can develop and build release artifacts that can be tested and released independently from artifacts for other projects. It offers various **advantages**, including improved team development and collaboration, facilitating automated testing and integration, and greatly reducing the need for manually tracking changes. **Dev Hub** must be enabled in the org to allow developers to create and manage scratch orgs and packages. In addition, to allow development using packages, **Enable Unlocked Packages and Second-Generation Managed Packages** must be selected on the Dev Hub page in Setup.



Requirement 2 (Part 1)



SCENARIO

The company uses a highly customized repository that contains a lot of different custom applications. For most of its Salesforce development needs, it is trying to shift from the org development model to the package development model. Currently, each developer works within their own personal sandbox. The architect needs to recommend a suitable development environment strategy for the package development model and provide considerations related to change tracking.



Requirement 2 (Part 2)



SCENARIO

Furthermore, the company recently became a Salesforce partner and would like to develop and launch a one-of-a-kind VR application that provides training to sales reps. It plans to use the org development model for developing the application. The architect needs to recommend a suitable development environment for this requirement.

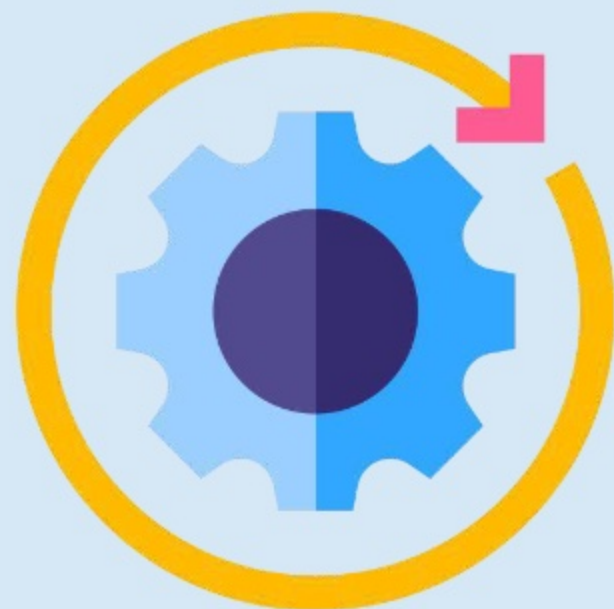


Solution 2 (Part 1)



SOLUTION

For development using the package development model, the company's developers should start using **scratch orgs**. A developer can spin up a scratch org to **start a new project** or **feature branch**, **test a new feature**, **perform automated testing**, etc. **Source Tracking** allows automatic tracking of changes between scratch orgs and Salesforce DX projects. It is **enabled by default** for scratch orgs. However, changes to metadata components **that are not yet supported** by source tracking need to be **tracked manually**.

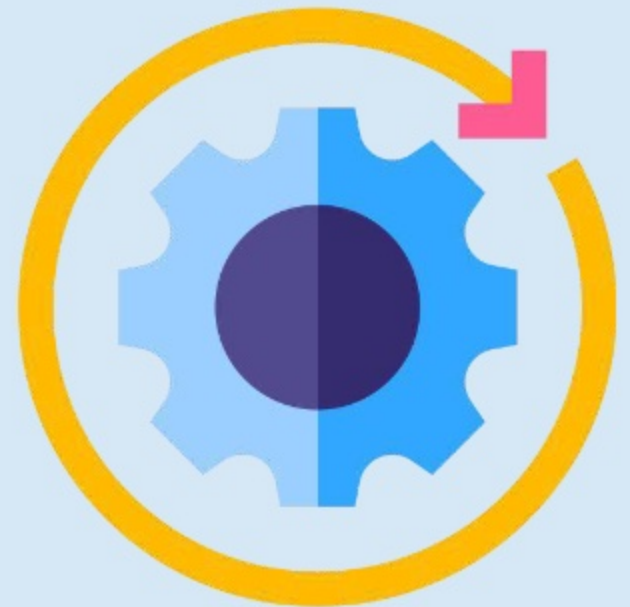


Solution 2 (Part 2)



SOLUTION

For the VR application, the company should use a **Partner Developer Edition org** for development. Compared to an individual Developer Edition org, it offers **higher API call limit**, **more storage**, and a **greater number of licenses**.



Learn More

- [!\[\]\(319cab28280b7a55668851c2b9d613f1_img.jpg\) Application Lifecycle and Development Models](#)
- [!\[\]\(3b323012153f0537b13ee8e42d8f4988_img.jpg\) Package Development Model](#)
- [!\[\]\(47842f0ce3f4b82d99e7f9521d86ee0a_img.jpg\) Org Development Model](#)
- [!\[\]\(324be7c9b450372d4141fc4412971e47_img.jpg\) Scratch Orgs](#)
- [!\[\]\(16b1353e9d6d2c9862dafd2ff716a08e_img.jpg\) Unlocked Packages](#)
- [!\[\]\(98028159d38692243b5c92b69d325c21_img.jpg\) Development Models](#)

